



US 20250280389A1

(19) **United States**

(12) **Patent Application Publication**
Li et al.

(10) **Pub. No.: US 2025/0280389 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **APPARATUSES AND METHODS FOR
ENHANCING HUB LOCATION
ESTIMATIONS USING CLOUD SOURCED
DATA**

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(21) Appl. No.: **19/209,896**

(22) Filed: **May 16, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/889,500, filed on
Aug. 17, 2022, now Pat. No. 12,335,906.

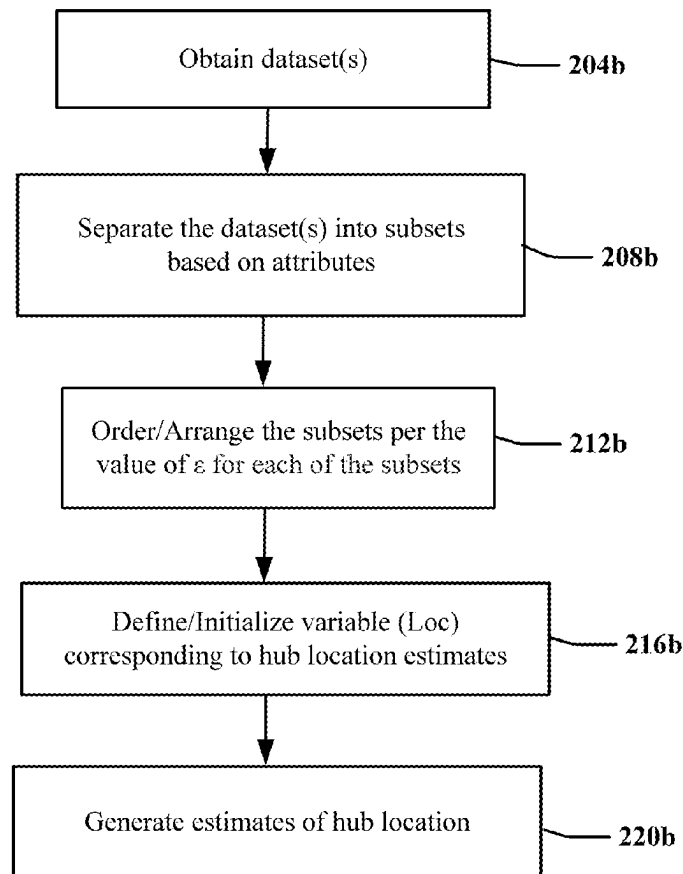
Publication Classification

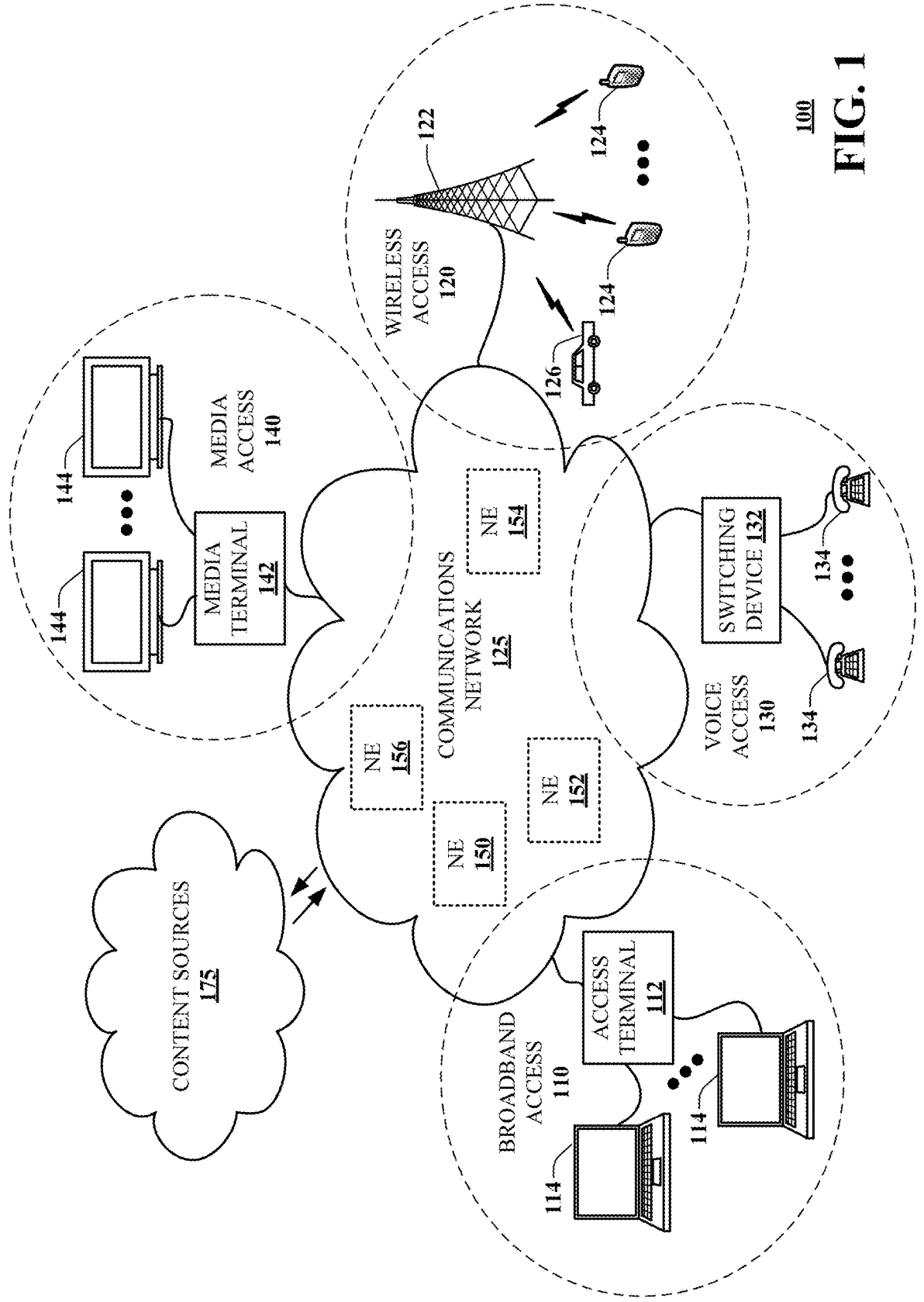
(51) **Int. Cl.**
H04W 64/00 (2009.01)

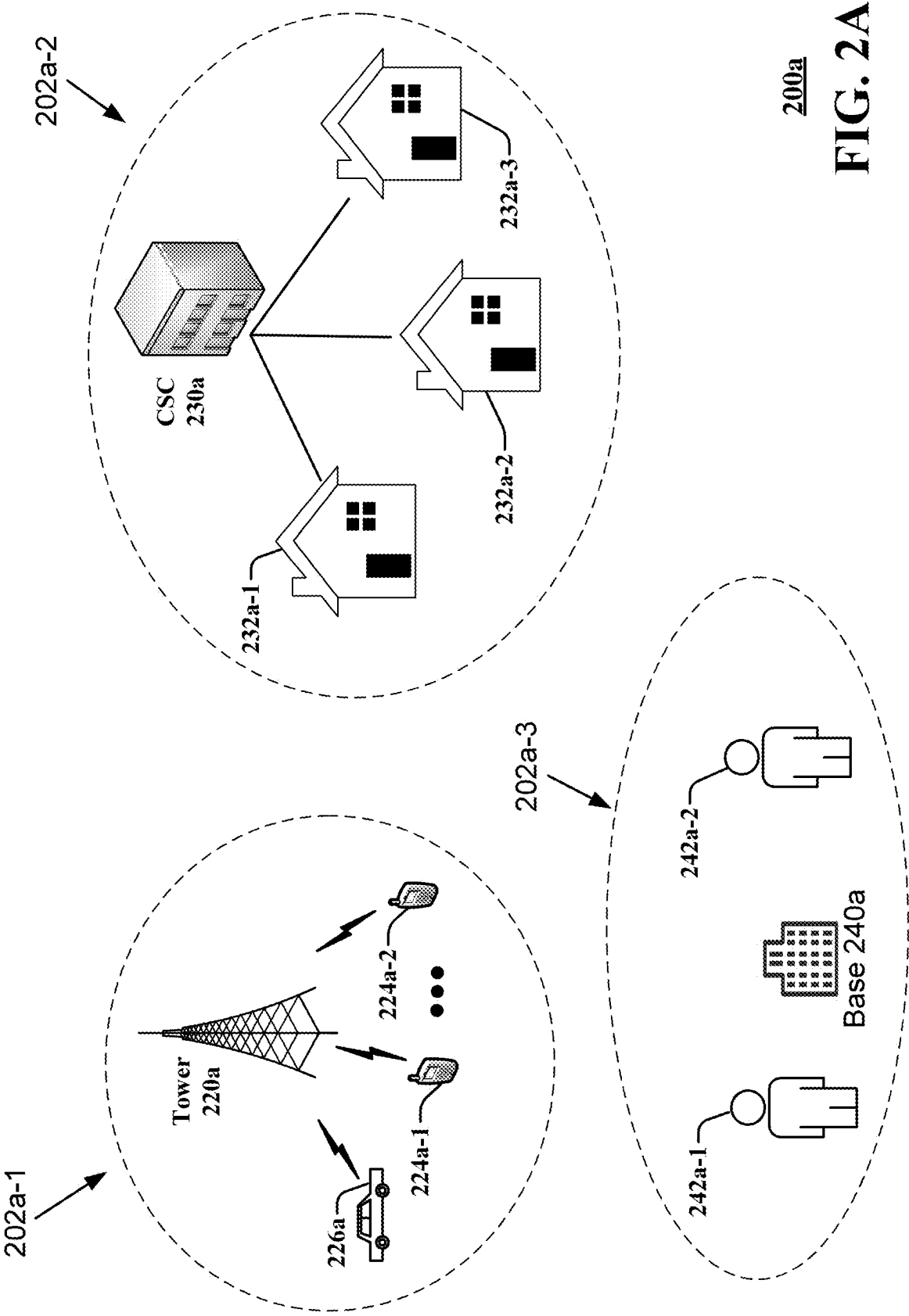
(52) **U.S. Cl.**
CPC **H04W 64/003** (2013.01)

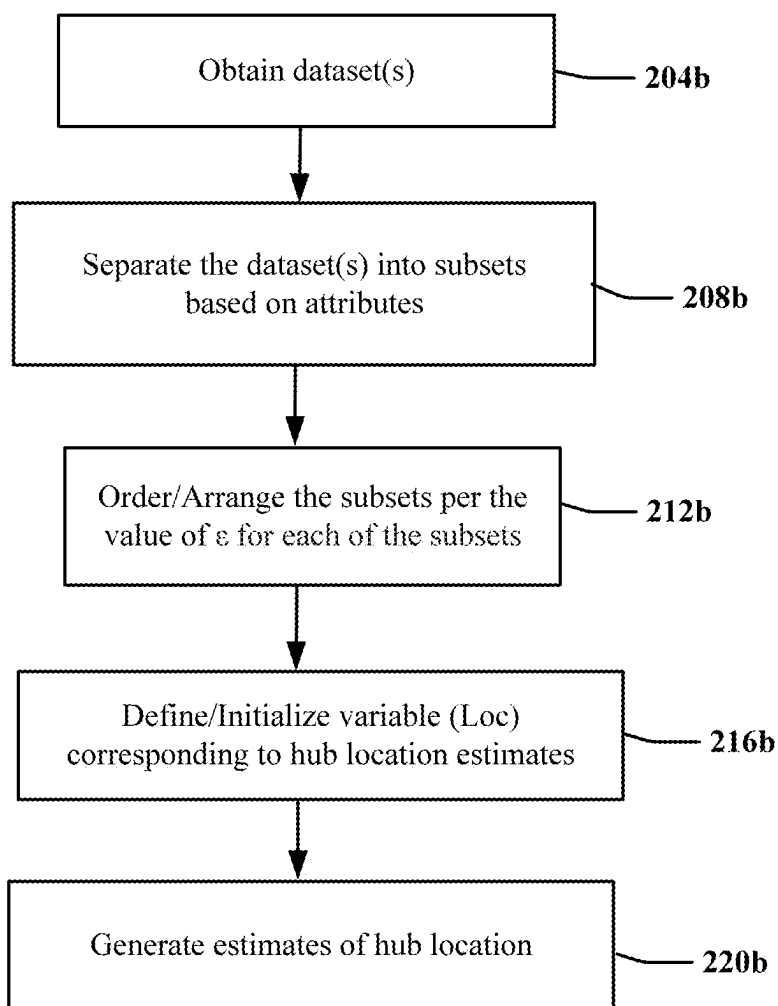
(57) **ABSTRACT**

Aspects of the subject disclosure may include, for example, obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of entities, separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a supported distance relative to a hub, and based on the ordering, generating an estimate of a location of the hub. Other embodiments are disclosed.









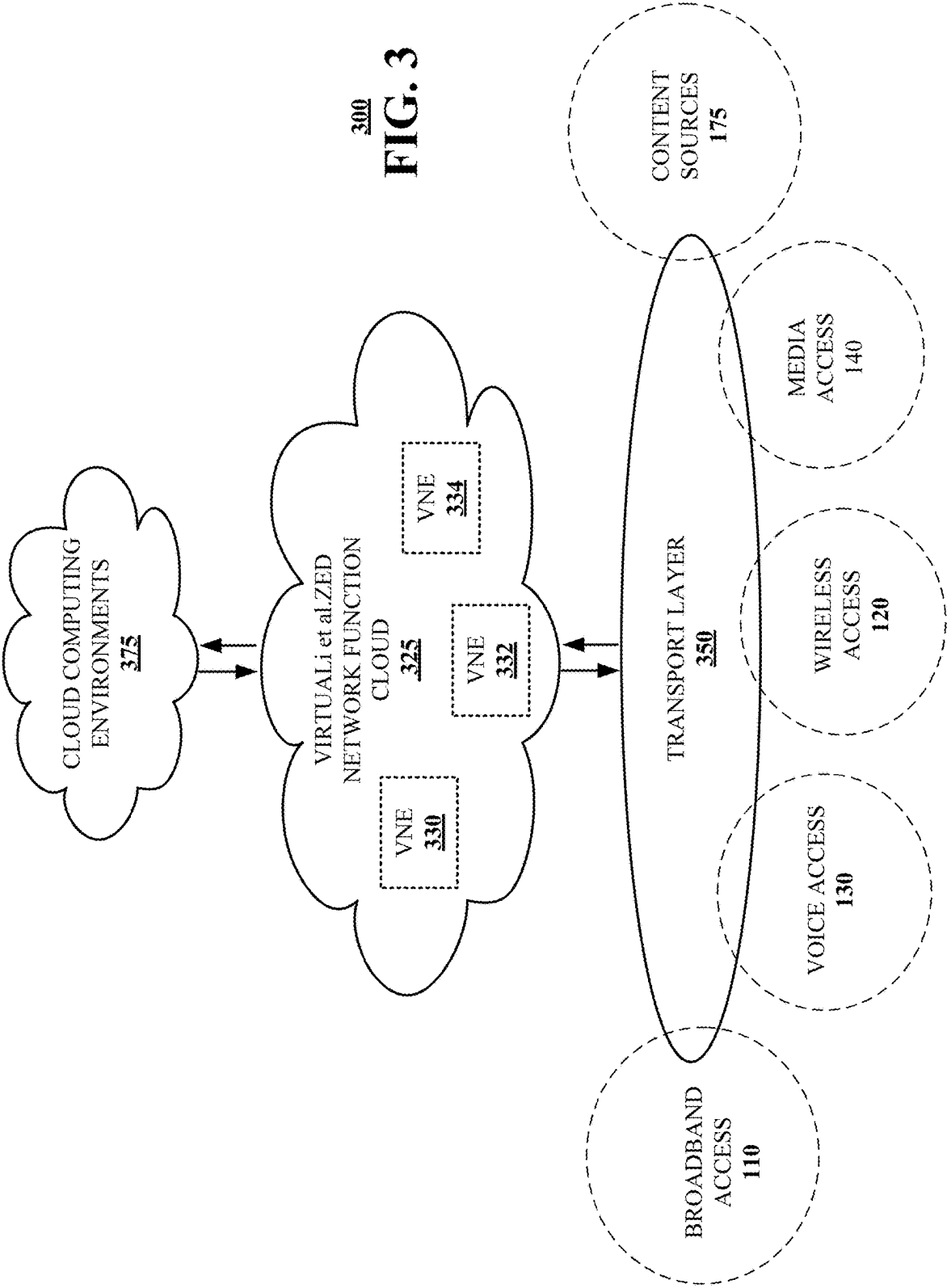
200b

FIG. 2B

```
for (t = 1; t++; t<=T) {  
    L = LocEST( $S_t$ ,  $\epsilon_t$ );  
    Loc = Loc + L;  
    for (k = t + 1; k++; k <= T) {  
         $S_k = R(S_k, L, \epsilon_k)$   
    }  
}
```

200c

FIG. 2C



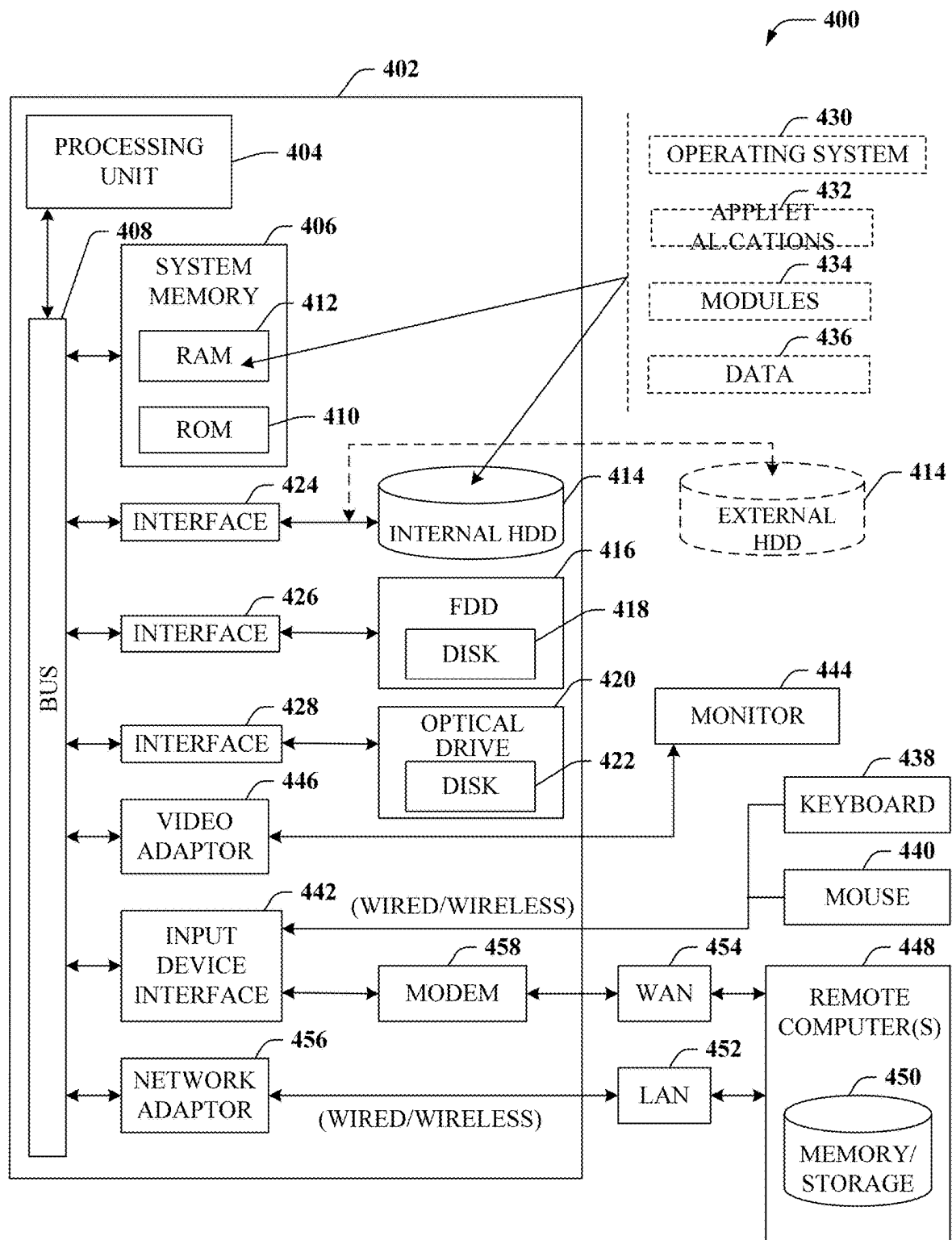


FIG. 4

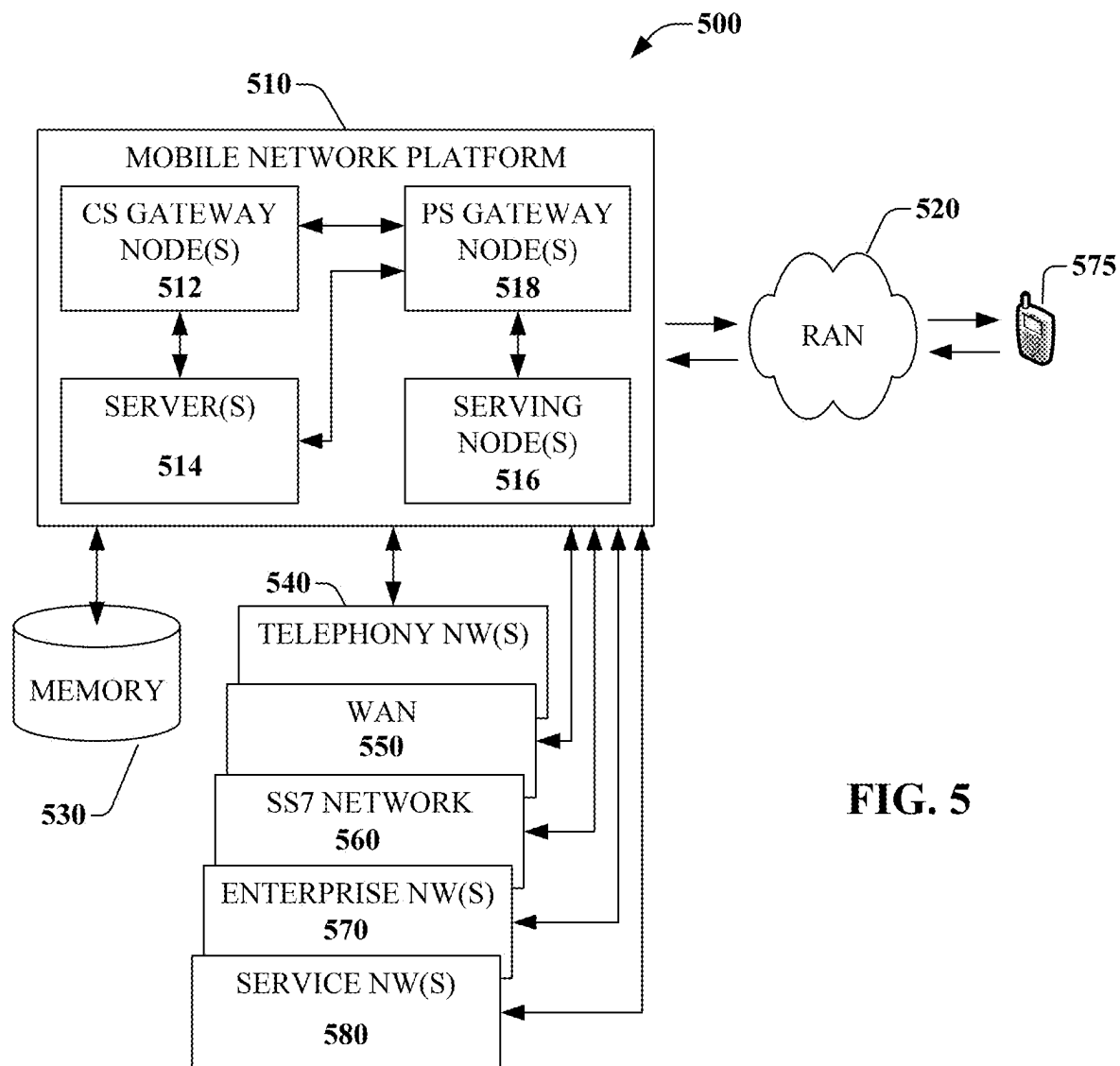
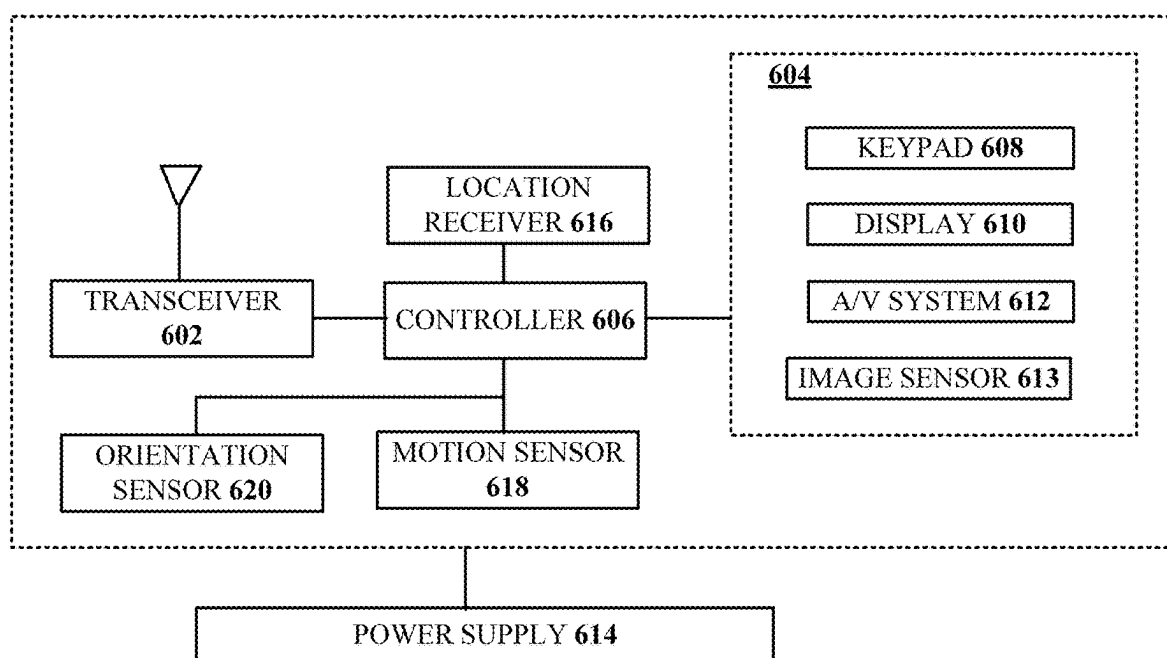


FIG. 5



600
FIG. 6

APPARATUSES AND METHODS FOR ENHANCING HUB LOCATION ESTIMATIONS USING CLOUD SOURCED DATA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/889,500, filed on Aug. 17, 2022. All sections of the aforementioned application are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The subject disclosure relates to apparatuses and methods for enhancing hub location estimations using cloud sourced data.

BACKGROUND

[0003] As the world becomes increasingly connected via vast communication networks and via various communication devices, additional challenges are imposed on network operators and service providers to ensure that sufficient resources are allocated or available to support communication services. For wireless network operators, knowing the locations of cell towers can be very useful in network maintenance, troubleshooting, competitive analysis, etc. This notion of a desire for knowing the location of cell towers may be extended to other practical applications pertaining to other types or kinds of resources. For example, in a military base setting it may be desirable to know the location of a base relative to a deployment/positioning of soldiers. In establishing a customer service/support center, a data source may pertain to a geographical location of target customer/subscriber residences, plus a tolerance related to maximum distance from the residences to the service center.

[0004] While a network operator/service provider usually has a record/log of its own tower locations, the record/log might not be accurate and current. For example, it may take some time from deployment until the presence of a new cell tower appears in the log. Similarly, if a tower is removed or decommissioned, time may be required before the removal/decommissioning is reflected in the log. Still further, and to support competitive analyses, a network operator/service provider would like to have insight into tower locations of other parties (such as competitors); however, this information can be difficult to attain. Still further, and in terms of a maximum distance parameter (e.g., a maximum distance relative to a location of a hub), there may exist attributes that may influence the value of the parameter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0006] FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

[0007] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system in accordance with various aspects described herein.

[0008] FIG. 2B depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0009] FIG. 2C illustrates code for identifying locations associated with a hub or a resource in accordance with aspects of this disclosure.

[0010] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0011] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0012] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0013] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0014] The subject disclosure describes, among other things, illustrative embodiments for enhancing accuracy in terms of a determination or estimate of a location of a resource or hub. Other embodiments are described in the subject disclosure.

[0015] One or more aspects of the subject disclosure include, in whole or in part, obtaining at least one dataset, the at least one dataset including first data pertaining to geographical information of a plurality of entities; separating the at least one dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to a different attribute of a plurality of attributes; ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub; and based on the ordering, estimating a location of the hub.

[0016] One or more aspects of the subject disclosure include, in whole or in part, obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of entities; separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes; ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub; and based on the ordering, generating an estimate of a location of the hub.

[0017] One or more aspects of the subject disclosure include, in whole or in part, parsing, by a processing system including a processor, a dataset to obtain a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes; ordering, by the processing system, the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub; and based on the ordering, generating, by the processing system, a respective estimate of a location of the hub for each subset of the plurality of subsets, resulting in a plurality of estimates.

[0018] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system 100 in accordance with various aspects described herein. For example, system 100 can facilitate in whole or in

part obtaining at least one dataset, the at least one dataset including first data pertaining to geographical information of a plurality of entities, separating the at least one dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to a different attribute of a plurality of attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, estimating a location of the hub. System **100** can facilitate in whole or in part obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of entities, separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating an estimate of a location of the hub. System **100** can facilitate in whole or in part parsing, by a processing system including a processor, a dataset to obtain a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering, by the processing system, the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating, by the processing system, a respective estimate of a location of the hub for each subset of the plurality of subsets, resulting in a plurality of estimates.

[0019] In particular, in FIG. 1 a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

[0020] The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VOIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0021] In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0022] In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0023] In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VOIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VOIP telephones and/or other telephony devices.

[0024] In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

[0025] In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0026] In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0027] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system **200a** in accordance with various aspects described herein. In some embodiments, one or more portions or aspects of the system **200a** may function within, or may be operatively overlaid upon, one or more portions of the system **100** of FIG. 1. FIG. 2A depicts the system **200a** in accordance with three different environments or practical applications represented by reference characters **202a-1**, **202a-2**, and **202a-3**. One skilled in the art will appreciate that the environments/applications **202a-1** through **202a-3** are illustrative, which is to say that aspects of this disclosure may be implemented or integrated within other environments/applications.

[0028] In the first environment/application **202a-1**, a tower **220a** is depicted. The tower **220a** may be used to

provide communication services in respect of devices or equipment, such as client devices/user equipment **224a-1** and **224a-2**, a connected vehicle **226a**, etc. In the second environment/application **202a-2**, a customer service center (CSC) **230a** is depicted. The CSC **230a** may be used to provide customer/subscriber support services for buildings (e.g., residences), such as buildings **232a-1**, **232a-2**, and **232a-3**. In the third environment/application **202a-3**, a base **240a** is depicted. The base **240a** may serve as a hub with respect to units of personnel, such as a first unit of personnel (represented by reference character **242a-1**) and a second unit of personnel (represented by reference character **242a-2**).

[0029] In various embodiments of the system **200a**, it may be desirable to know the locations of the tower **220a**, the CSC **230a**, and/or the base **240a**. For example, given a dataset of geographical/geo information of hub linking entities (e.g., such as a dataset pertaining to the client devices **224a-1** and **224a-2**, the connected car **226a**, the buildings **232a-1** through **232a-3**, and/or the units **242a-1** and **242a-2**), it may be desirable to estimate the hub locations (e.g., the locations of the tower **220a**, the CSC **230a**, and/or the base **240a**) as accurately as possible. Techniques for estimating hub locations using source data with geo information will be appreciated by one of skill in the art. U.S. patent application Ser. No. 17/489,834, filed on Sep. 30, 2021, discloses examples of such techniques; the contents of U.S. patent application Ser. No. 17/489,834 are incorporated herein by reference in their entirety.

[0030] An estimation error, in terms of the location of a hub, may be expressed or defined as the distance between an estimated location of a hub and an actual location of the hub. Given a maximum hub-to-entity distance parameter (ϵ), the larger the value of ϵ , the greater the estimation error.

[0031] In an effort to reduce the estimation error (e.g., in order to obtain greater accuracy in respect of an estimated location of a hub), additional data/datasets, such as cloud sourced data/datasets pertaining to attributes that can impact the value of, may be utilized. For example, in a tower-based application (see **202a-1** in FIG. 2A), a type of radio access network (RAN) technology (e.g., 5G millimeter Wave, 5G Sub6 (e.g., below 6 GHz frequency), LTE, etc.) may impact the maximum distance from a device (e.g., user equipment/client device) to a tower. In a customer service/support center scenario (see **202a-2** in FIG. 2A), different types of customers/subscribers (e.g., whether a customer/subscriber owns a car or not) may tolerate or support different tolerances in terms of a (maximum) distance parameter value relative to a CSC (e.g., different communication services may have an associated maximum value in terms of round-trip delay). In a military-based application (see **202a-3** in FIG. 2A), different types of units of personnel/soldiers (e.g., special units vs. regular units) may tolerate or support a different (maximum) distance relative to the base.

[0032] Referring now to FIG. 2B, an illustrative embodiment of a method **200b** in accordance with various aspects described herein is shown. The method **200b** may be implemented (e.g., executed), in whole or in part, in conjunction with one or more systems, devices, and/or components, such as for example the systems, devices, and components set forth herein. The method **200b** may be implemented to facilitate an identification/determination or estimate of one or more locations of one or more hubs.

[0033] The method **200b** is described below in relation to the code **200c** shown in FIG. 2C. The code **200c** is illustrative, which is to say that aspects of the method **200b** may be implemented using code that is different from what is shown in FIG. 2C. More generally, aspects of the method **200b** may be implemented using hardware, software, firmware, or any combination thereof. The code **200c** may be implemented via instructions that may be executed by a processing system including one or more processors. Operations associated with the code are described below in relation to the blocks of FIG. 2B.

[0034] Before proceeding with a fuller description of the method **200b**, it is noted that the code **200c** makes use of two functions. A first of the functions (1) may be expressed as:

$$L = \text{LocEST}(S, \epsilon)$$

[0035] where LocEST corresponds to techniques/methods for operating on a dataset S and parameter ϵ to produce an estimated hub location set L . Again, reference may be made to the aforementioned U.S. patent application Ser. No. 17/489,834 for examples of the use of such techniques/methods.

[0036] A second of the functions (2) may be expressed as:

$$S = R(S_0, L, \epsilon)$$

[0037] where the function R corresponds to a removal of datapoints within a distance (e.g., a radius) ϵ of location set L from dataset S_0 , producing/resulting in a reduced dataset S .

[0038] In block **204b**, one or more datasets may be obtained from one or more sources. The datasets may include, or be based on, geo information and/or attribute information.

[0039] In block **208b**, the datasets obtained as part of block **204b** may be separated/parsed into subsets with different attributes (or, analogously, different groups of attributes), each with its own respective parameter ϵ . To the extent that a subset includes multiple/plural attributes, the parameter ϵ that is used for the subset may correspond to a maximum ϵ amongst the respective instances/values of ϵ for the subset.

[0040] In block **212b**, the subsets of block **208b** may be ordered or arranged in accordance with an increasing value of ϵ , so that dataset $S = S_1 + S_2 + \dots + S_T$ and $\epsilon_1 < \epsilon_2 < \dots < \epsilon_T$, with $T > 1$ and T corresponding to the count or number of subsets. Thus, S_i corresponds to the data subset for attribute i and ϵ_i corresponds to the ϵ parameter for data subset S_i , where $i = [1, T]$.

[0041] In block **216b**, a variable (Loc) may be defined and/or initialized to an empty location set. For reasons that will become clearer below, the variable Loc may be used to store values corresponding to hub location estimates.

[0042] In block **220b**, the code **200c** may be executed to generate an estimate of the hub locations. The code **200c** is arranged as a pair of (nested) for loops. The outer for loop (e.g., the for loop defined by the parameters: $t=1$; $t++$; $t \leq T$ in FIG. 2C) may iterate upon t to obtain the location estimate (LocEST) for the given value of t ; the location estimate corresponding to the value of t may be added to the variable Loc. The inner for loop (e.g., the for loop defined by the parameters: $k=t+1$; $k++$; $k \leq T$ in FIG. 2C) may iterate upon k to remove (R) data points within the radius ϵ_k of location set L from the dataset S_k to produce a reduced dataset S_k . The use of the inner for loop may be based on a recognition that data outside of a maximum distance/range ϵ may be

removed/eliminated. Following the execution of block **220b**, the variable *Loc* may contain values corresponding to the hub location estimates. To the extent that the estimates for a given hub differ from one another, one or more filtering or averaging techniques may be applied as part of block **220b** to generate an overall estimate.

[0043] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2B, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0044] As set forth herein, aspects of this disclosure may be included/integrated as part of one or more practical applications. For example, aspects of this disclosure may serve to generate an estimate of a location of a hub (or, analogously, a location of a resource), which can be useful in many practical applications involving communications, positioning of assets or personnel, and the like. Based on a use of attribute-driven data, an accuracy of the hub location estimate may be enhanced with the understanding that attributes may influence/impact the (maximum) distance parameter ϵ discussed above. Thus, aspects of this disclosure represent substantial improvements vis-à-vis prior techniques for estimating locations.

[0045] As set forth above, aspects of this disclosure may be used to generate useful, concrete, and tangible results. Aspects of this disclosure may be tied to (specially or specifically programmed) machines or apparatuses that may be configured to generate the aforementioned results. Furthermore, aspects of this disclosure are transformative in nature, as data pertaining to attributes associated with an environment or application may be adapted to correspond to a location of a hub or resource. In brief, as one of skill in the art will appreciate based on a review of this disclosure, aspects of this disclosure are not directed to abstract ideas. To the contrary, aspects of this disclosure are directed to, and encompass, substantially more than any abstract idea standing alone.

[0046] Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of system **200a**, and method **200b** presented in FIGS. 1, 2A, and 2B. For example, virtualized communication network **300** can facilitate in whole or in part obtaining at least one dataset, the at least one dataset including first data pertaining to geographical information of a plurality of entities, separating the at least one dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to a different attribute of a plurality of attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, estimating a location of the hub. Virtualized communication network **300** can facilitate in whole or in part obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of

entities, separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating an estimate of a location of the hub. Virtualized communication network **300** can facilitate in whole or in part parsing, by a processing system including a processor, a dataset to obtain a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering, by the processing system, the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating, by the processing system, a respective estimate of a location of the hub for each subset of the plurality of subsets, resulting in a plurality of estimates.

[0047] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0048] In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general-purpose processors or general-purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0049] As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it is elastic: so, the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle boxes are instantiated from the common resource pool. Such

sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0050] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized and might require special DSP code and analog front ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0051] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements do not typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and which creates an elastic function with higher availability overall than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud computing services.

[0052] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud or might simply orchestrate workloads supported entirely in NFV infrastructure from these third-party locations.

[0053] Turning now to FIG. 4, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. 4 and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs

330, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part obtaining at least one dataset, the at least one dataset including first data pertaining to geographical information of a plurality of entities, separating the at least one dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to a different attribute of a plurality of attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, estimating a location of the hub. Computing environment **400** can facilitate in whole or in part obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of entities, separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating an estimate of a location of the hub. Computing environment **400** can facilitate in whole or in part parsing, by a processing system including a processor, a dataset to obtain a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering, by the processing system, the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating, by the processing system, a respective estimate of a location of the hub for each subset of the plurality of subsets, resulting in a plurality of estimates.

[0054] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0055] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0056] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communica-

tions network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0057] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0058] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0059] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0060] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0061] With reference again to FIG. 4, the example environment can comprise a computer 402, the computer 402 comprising a processing unit 404, a system memory 406 and a system bus 408. The system bus 408 couples system components including, but not limited to, the system memory 406 to the processing unit 404. The processing unit 404 can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit 404.

[0062] The system bus 408 can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory 406 comprises ROM 410 and RAM 412. A basic input/output system (BIOS) can

be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer 402, such as during startup. The RAM 412 can also comprise a high-speed RAM such as static RAM for caching data.

[0063] The computer 402 further comprises an internal hard disk drive (HDD) 414 (e.g., EIDE, SATA), which internal HDD 414 can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) 416, (e.g., to read from or write to a removable diskette 418) and an optical disk drive 420, (e.g., reading a CD-ROM disk 422 or, to read from or write to other high-capacity optical media such as the DVD). The HDD 414, magnetic FDD 416 and optical disk drive 420 can be connected to the system bus 408 by a hard disk drive interface 424, a magnetic disk drive interface 426 and an optical drive interface 428, respectively. The hard disk drive interface 424 for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0064] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer 402, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0065] A number of program modules can be stored in the drives and RAM 412, comprising an operating system 430, one or more application programs 432, other program modules 434 and program data 436. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM 412. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0066] A user can enter commands and information into the computer 402 through one or more wired/wireless input devices, e.g., a keyboard 438 and a pointing device, such as a mouse 440. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit 404 through an input device interface 442 that can be coupled to the system bus 408, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0067] A monitor 444 or other type of display device can be also connected to the system bus 408 via an interface, such as a video adapter 446. It will also be appreciated that in alternative embodiments, a monitor 444 can also be any

display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer 402 via any communication means, including via the Internet and cloud-based networks. In addition to the monitor 444, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0068] The computer 402 can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) 448. The remote computer(s) 448 can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer 402, although, for purposes of brevity, only a remote memory/storage device 450 is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) 452 and/or larger networks, e.g., a wide area network (WAN) 454. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0069] When used in a LAN networking environment, the computer 402 can be connected to the LAN 452 through a wired and/or wireless communication network interface or adapter 456. The adapter 456 can facilitate wired or wireless communication to the LAN 452, which can also comprise a wireless AP disposed thereon for communicating with the adapter 456.

[0070] When used in a WAN networking environment, the computer 402 can comprise a modem 458 or can be connected to a communications server on the WAN 454 or has other means for establishing communications over the WAN 454, such as by way of the Internet. The modem 458, which can be internal or external and a wired or wireless device, can be connected to the system bus 408 via the input device interface 442. In a networked environment, program modules depicted relative to the computer 402 or portions thereof, can be stored in the remote memory/storage device 450. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0071] The computer 402 can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0072] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A

Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0073] Turning now to FIG. 5, an embodiment 500 of a mobile network platform 510 is shown that is an example of network elements 150, 152, 154, 156, and/or VNEs 330, 332, 334, etc. For example, platform 510 can facilitate in whole or in part obtaining at least one dataset, the at least one dataset including first data pertaining to geographical information of a plurality of entities, separating the at least one dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to a different attribute of a plurality of attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, estimating a location of the hub. Platform 510 can facilitate in whole or in part obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of entities, separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating an estimate of a location of the hub. Platform 510 can facilitate in whole or in part parsing, by a processing system including a processor, a dataset to obtain a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering, by the processing system, the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating, by the processing system, a respective estimate of a location of the hub for each subset of the plurality of subsets, resulting in a plurality of estimates.

[0074] In one or more embodiments, the mobile network platform 510 can generate and receive signals transmitted and received by base stations or access points such as base station or access point 122. Generally, mobile network platform 510 can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform 510 can be included in telecommunications carrier networks and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform 510 comprises CS gateway node(s) 512 which can interface CS traffic received from legacy networks like telephony network(s) 540 (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system

#7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

[0075] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0076] In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0077] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise

utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. 1(s) that enhance wireless service coverage by providing more network coverage.

[0078] It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0079] In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0080] In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0081] Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part obtaining at least one dataset, the at least one dataset including first data pertaining to geographical information of

a plurality of entities, separating the at least one dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to a different attribute of a plurality of attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, estimating a location of the hub. Computing device 600 can facilitate in whole or in part obtaining a dataset, the dataset including first data pertaining to geographical information associated with a plurality of entities, separating the dataset into a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating an estimate of a location of the hub. Computing device 600 can facilitate in whole or in part parsing, by a processing system including a processor, a dataset to obtain a plurality of subsets, wherein each subset of the plurality of subsets corresponds to at least one attribute of a plurality of different attributes, ordering, by the processing system, the plurality of subsets in accordance with respective parameter values associated with each of the subsets, wherein each of the respective parameter values corresponds to a maximum supported distance relative to a hub, and based on the ordering, generating, by the processing system, a respective estimate of a location of the hub for each subset of the plurality of subsets, resulting in a plurality of estimates.

[0082] The communication device 600 can comprise a wireline and/or wireless transceiver 602 (herein transceiver 602), a user interface (UI) 604, a power supply 614, a location receiver 616, a motion sensor 618, an orientation sensor 620, and a controller 606 for managing operations thereof. The transceiver 602 can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, Wi-Fi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver 602 can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VOIP, etc.), and combinations thereof.

[0083] The UI 604 can include a depressible or touch-sensitive keypad 608 with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device 600. The keypad 608 can be an integral part of a housing assembly of the communication device 600 or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad 608 can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI 604 can further include a display 610 such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic

Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device 600. In an embodiment where the display 610 is touch-sensitive, a portion or all of the keypad 608 can be presented by way of the display 610 with navigation features.

[0084] The display 610 can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device 600 can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display 610 can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display 610 can be an integral part of the housing assembly of the communication device 600 or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0085] The UI 604 can also include an audio system 612 that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation). The audio system 612 can further include a microphone for receiving audible signals of an end user. The audio system 612 can also be used for voice recognition applications. The UI 604 can further include an image sensor 613 such as a charged coupled device (CCD) camera for capturing still or moving images.

[0086] The power supply 614 can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device 600 to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0087] The location receiver 616 can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device 600 based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor 618 can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device 600 in three-dimensional space. The orientation sensor 620 can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device 600 (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

[0088] The communication device 600 can use the transceiver 602 to also determine a proximity to a cellular, Wi-Fi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller 606 can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, applica-

tion specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device 600.

[0089] Other components not shown in FIG. 6 can be used in one or more embodiments of the subject disclosure. For instance, the communication device 600 can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

[0090] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0091] In the subject specification, terms such as “store,” “storage,” “data store,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, non-volatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-DRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0092] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0093] In one or more embodiments, information regarding use of services can be generated including services being

accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0094] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $x=(x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence (class)}$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0095] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to pre-determined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

[0096] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable,

a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0097] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0098] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should

generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0099] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” “subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0100] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0101] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0102] As used herein, terms such as “data storage,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0103] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is

intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0104] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0105] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0106] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

What is claimed is:

1. A device, comprising:

a processing system including a processor; and

a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:

obtaining a plurality of datasets from at least one source; parsing the plurality of datasets into a plurality of subsets, wherein each subset of the plurality of subsets has different attributes relative to the other subsets of the plurality of subsets;

selecting, for each subset of the plurality of subsets, a parameter for the subset corresponding to a maximum value amongst respective values of the parameter; ordering, based on the selecting, the plurality of subsets; and

estimating, based on the ordering, a location of a hub.

2. The device of claim 1, wherein the hub corresponds to first network infrastructure.

3. The device of claim 2, wherein the first network infrastructure comprises: a cellular tower, a base station, a switch, a router, a gateway, a server, or any combination thereof.

4. The device of claim 1, wherein the plurality of datasets corresponds to a plurality of entities.

5. The device of claim 4, wherein the plurality of entities comprises a mobile device, a connected vehicle, or a combination thereof.

6. The device of claim 4, wherein the hub corresponds to a base of operations for units of personnel included in the plurality of entities.

7. The device of claim 6, wherein the base of operations comprises a military base, and wherein the units of personnel comprise a special forces unit and a regular unit.

8. The device of claim 4, wherein the hub corresponds to a customer service center for supporting communication services.

9. The device of claim 8, wherein the plurality of entities comprises a plurality of residences supported by the customer service center.

10. The device of claim 1, wherein the respective parameter for each subset of the plurality of subsets is based on a supported distance relative to the hub.

11. The device of claim 10, wherein the supported distance is expressed as a function of a maximum round-trip delay tolerated by a respective communication service supported by the subset.

12. The device of claim 1, wherein each attribute of the different attributes corresponds to a type of a plurality of types of radio access network (RAN) technologies.

13. The device of claim 12, wherein the plurality of types comprises: 5G millimeter Wave, 5G Sub6, and LTE.

14. The device of claim 1, wherein the estimating of the location of the hub comprises applying a filtering or averaging technique to a plurality of estimates of the location of the hub to generate an overall estimate for the location of the hub.

15. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising:

obtaining a plurality of datasets from at least one source; parsing the plurality of datasets into a plurality of subsets, wherein each subset of the plurality of subsets has at least one different attribute relative to the other subsets of the plurality of subsets;

selecting, for each subset of the plurality of subsets, a parameter for the subset corresponding to a maximum value amongst respective values of the parameter;

ordering, based on the selecting, the plurality of subsets;
and

estimating, based on the ordering, a location of a hub.

16. The non-transitory machine-readable medium of claim **15**, wherein the estimating is based on an iterative determination of the location of the hub in accordance with each subset of the plurality of subsets.

17. The non-transitory machine-readable medium of claim **16**, wherein the iterative determination comprises a removal of data from the subsets that are associated with a range that exceeds a supported distance for the subset.

18. A method, comprising:

parsing, by a processing system including a processor, a plurality of datasets into a plurality of subsets, wherein each subset of the plurality of subsets has different attributes relative to the other subsets of the plurality of subsets;

selecting, by the processing system and for each subset of the plurality of subsets, a parameter for the subset corresponding to a maximum value amongst respective values of the parameter;

ordering, by the processing system and based on the selecting, the plurality of subsets; and

estimating, by the processing system and based on the ordering, a location of a hub, resulting in an estimated location of the hub.

19. The method of claim **18**, further comprising:

determining, by the processing system, an actual location of the hub.

20. The method of claim **19**, further comprising:

generating, by the processing system, a value of error corresponding to a difference between the estimated location of the hub and the actual location of the hub.

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